1950-2024 / REVISED ANALYSIS

Formula 1 World Championship

A system of drivers, teams and race outcomes.

An exploratory study of performance, qualifying, pit stops and teammate networks.



PRESENTER

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Source snapshot: Kaggle version 24 | Analysis reviewed: 08 Nov 2026

Data, scope and analytical method

26,759
RACE-RESULT ROWS

1,125
RACES

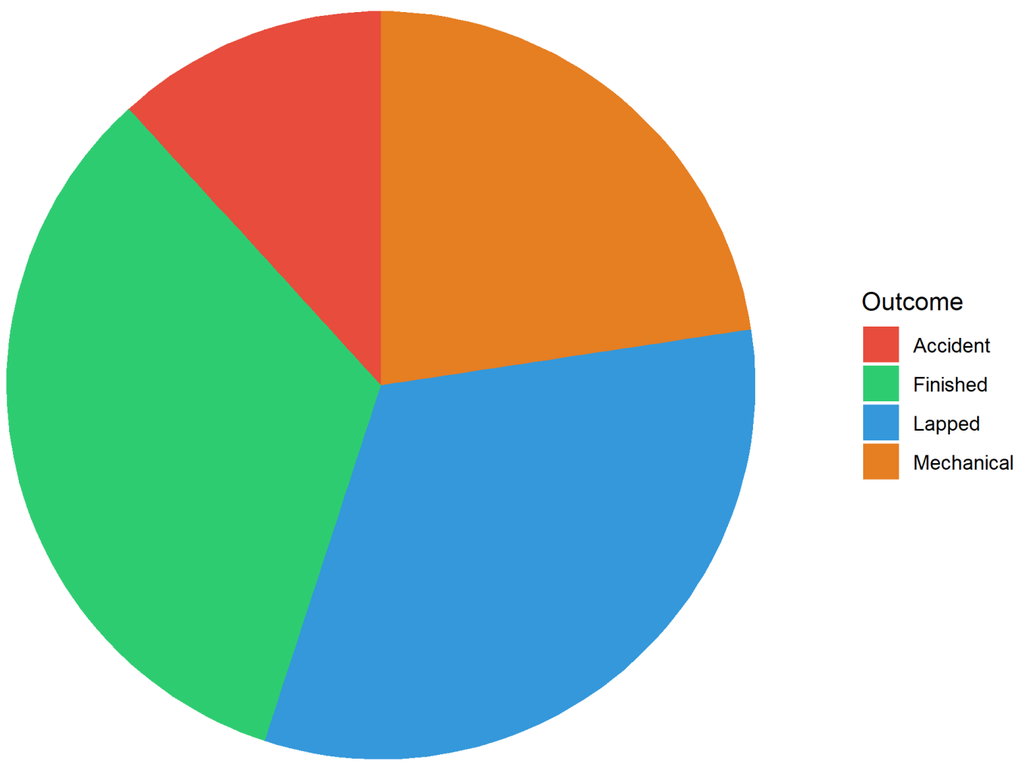
861
DRIVERS

Eight CSV tables connect drivers, constructors, races, circuits, outcomes, qualifying and pit stops. The study uses races from 1950 through 2024.

Method: validate IDs; join lookup tables without changing result-row counts; calculate descriptive summaries; build unique teammate edges; export plots; check output invariants.

Boundaries: no causal inference, no inference from surnames, and no street/permanent classification inferred from countries. An optional reviewed circuit-ID mapping can enable those comparisons.

How do race-result entries end?



DEFINITION

Categories are an analytical grouping of recorded result statuses.

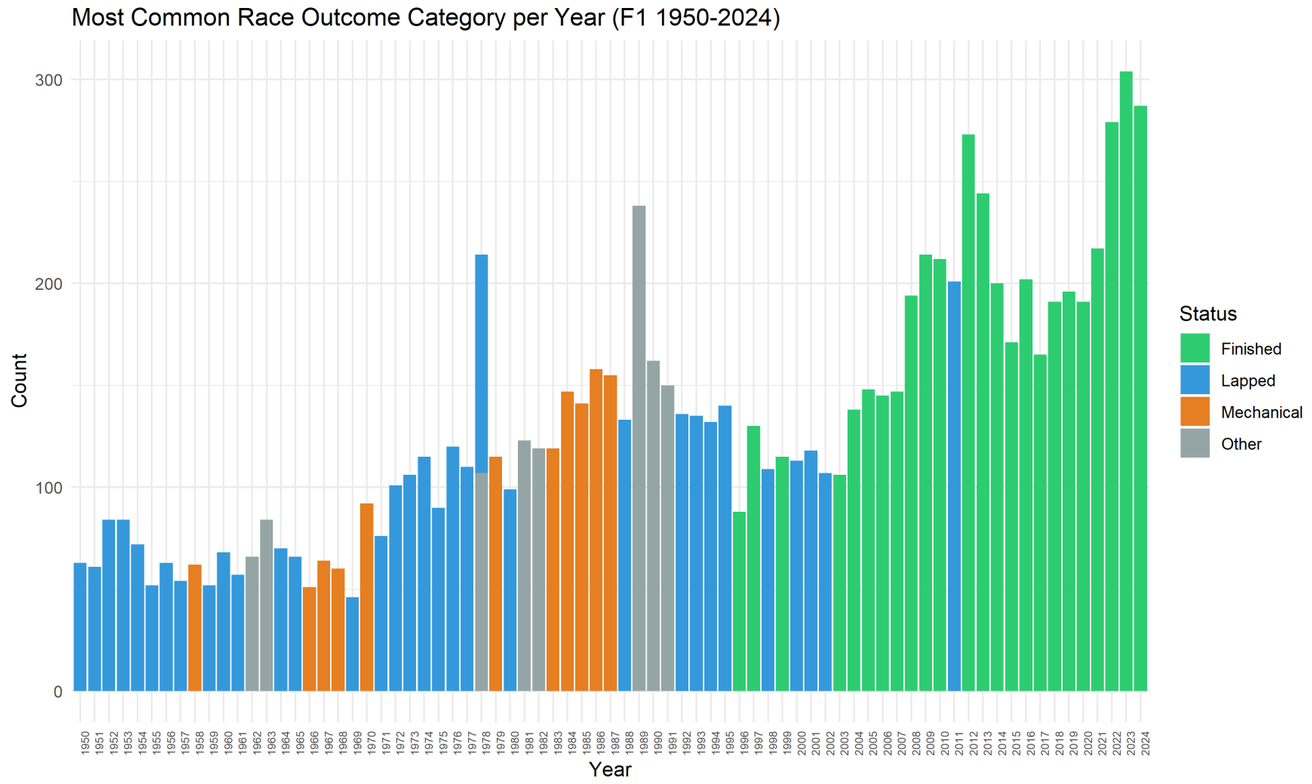
SCOPE

This pie excludes Other. It describes only the four displayed categories.

CORRECTION

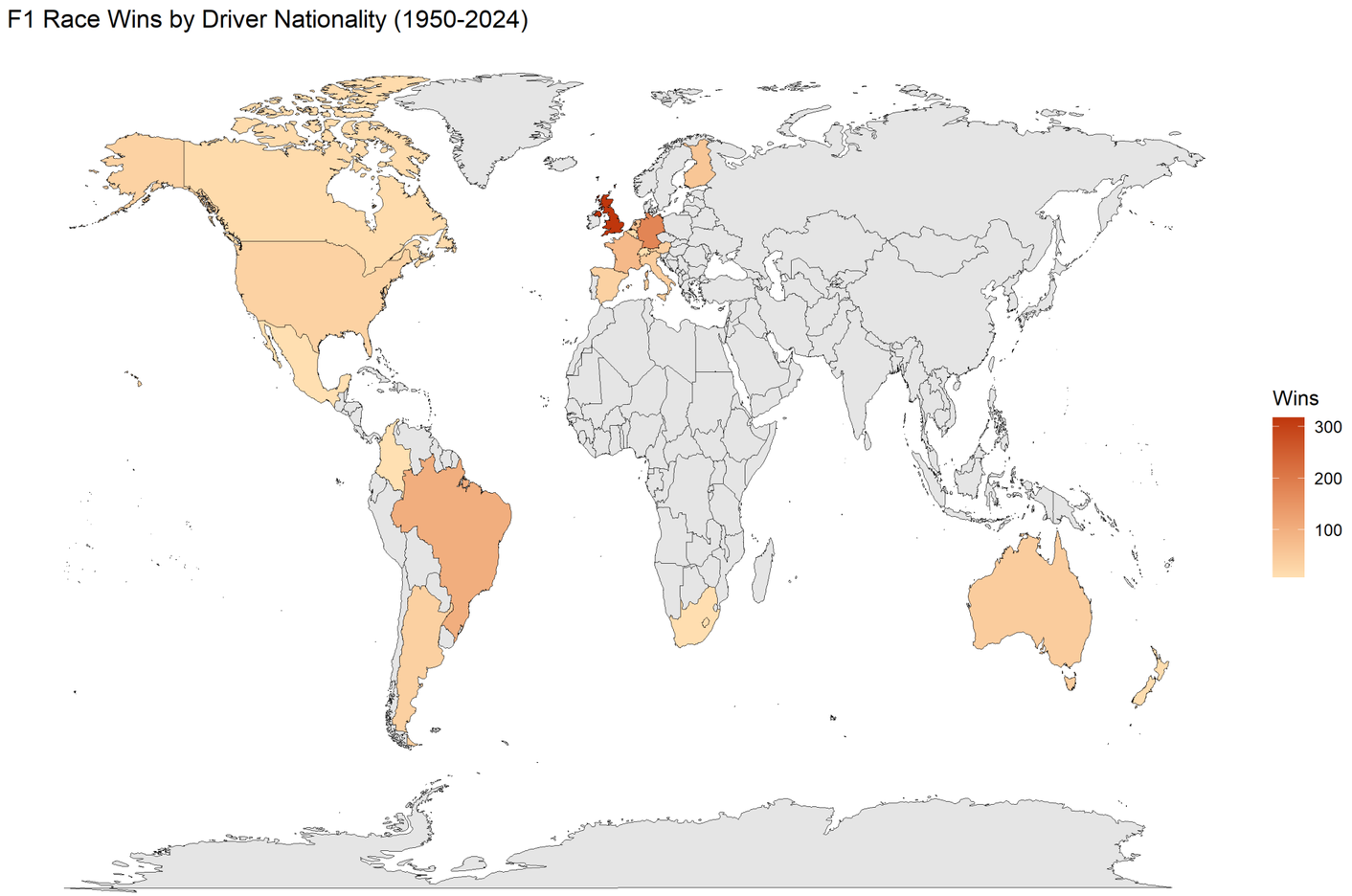
A generic Retired status is grouped as Other; it is not evidence of an accident.

Which outcome dominates each season?



Each bar reports the count for a season's most frequent category; ties can produce multiple bars. Counts are not rates and are affected by the number of races and entries.

Where do race winners come from?



MEASURE

Wins grouped by the recorded nationality of each winning driver.

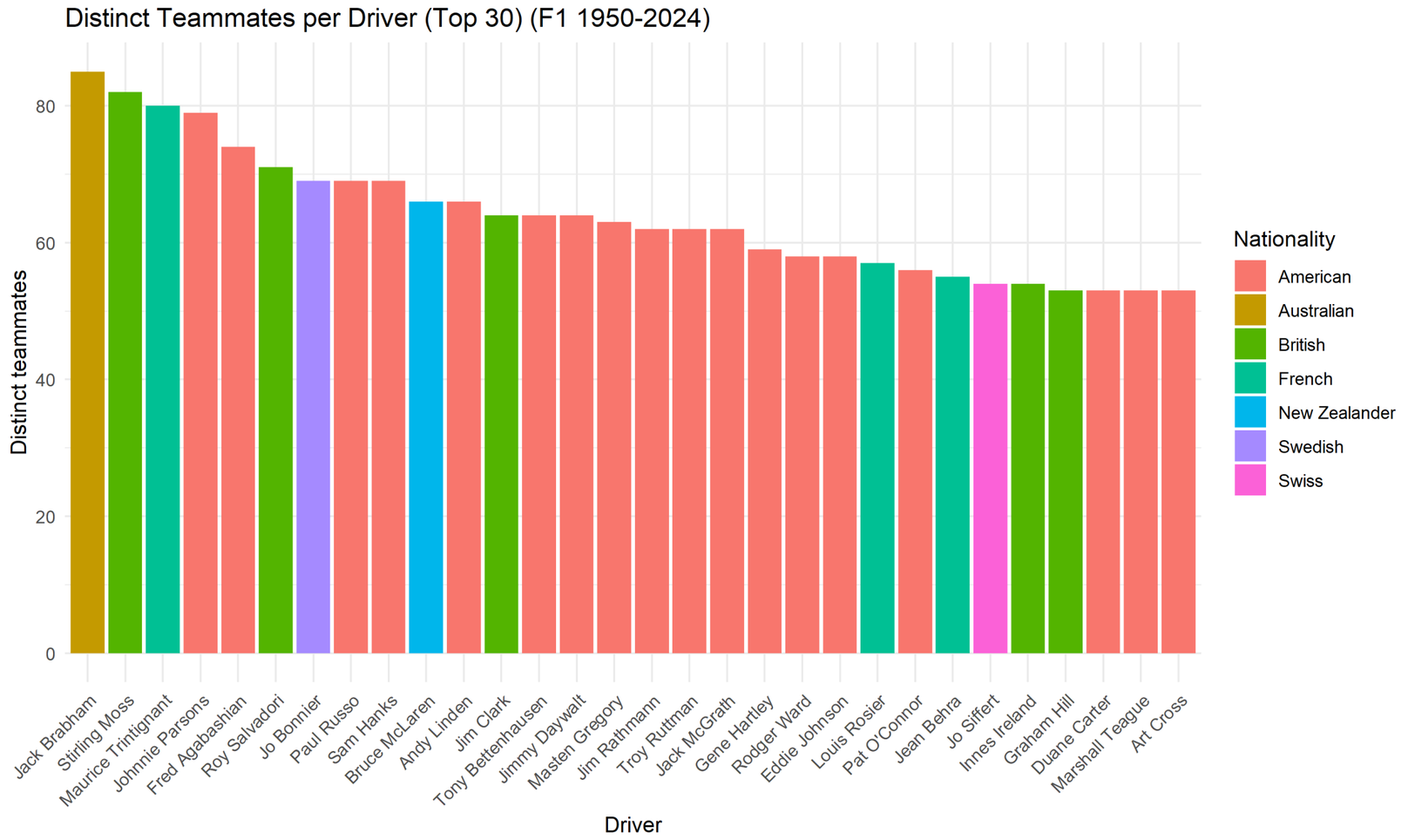
INTERPRETATION

The map describes accumulated wins, not a country's causal influence on performance.

LIMIT

Historical nationality labels and country-map matching affect geographic coverage.

Drivers connected through shared teams



IDENTITY

Driver IDs distinguish different people who share a surname.

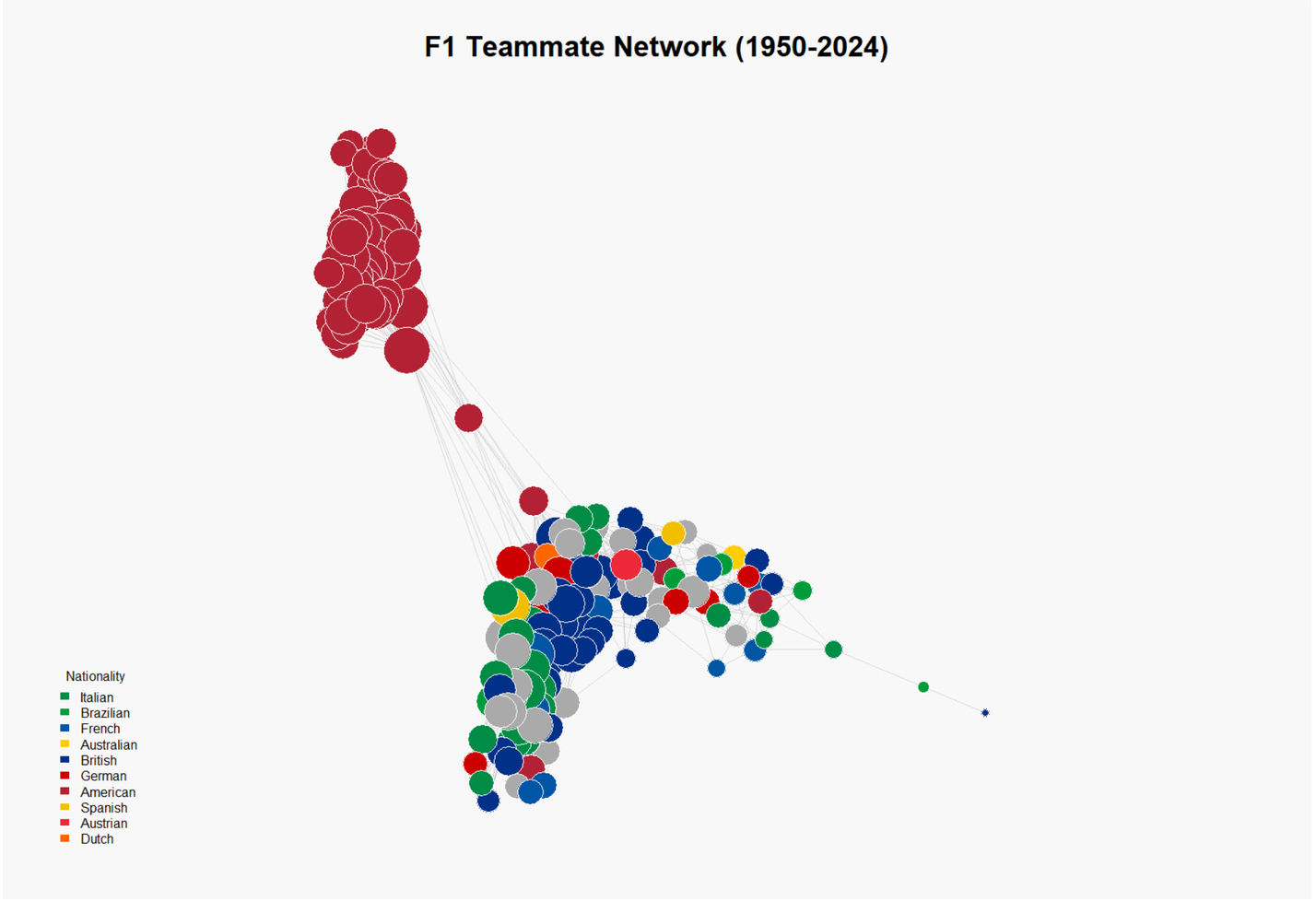
MEASURE

A pairing is a distinct teammate, not the number of races the pair contested together.

DISPLAY

The top 30 drivers are shown. Historic shared-car and team-entry records affect the definition.

The teammate network



FULL GRAPH

815 connected drivers and 5,006 unique undirected edges.

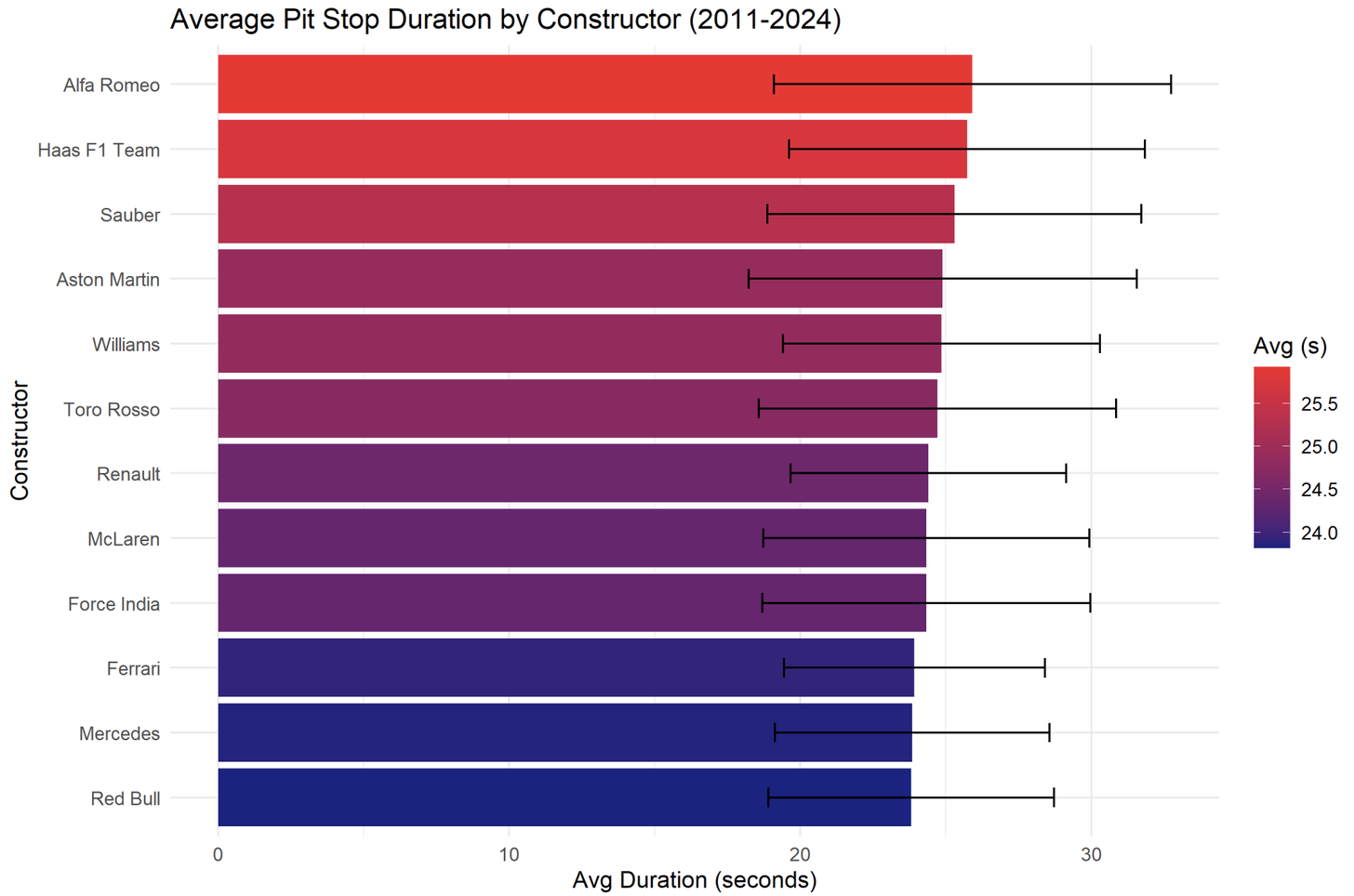
DISPLAYED GRAPH

Only drivers with at least 15 distinct teammates in the full graph are displayed. Labels are omitted for clarity; IDs and names are available in the node table.

BOUNDARY

46 drivers have no teammate edge under this definition and are absent from the graph. This filter does not remove Indianapolis 500 entries.

Recorded pit-stop durations



SELECTION

The 12 constructors with the most recorded stops are compared after filtering durations below 120 seconds.

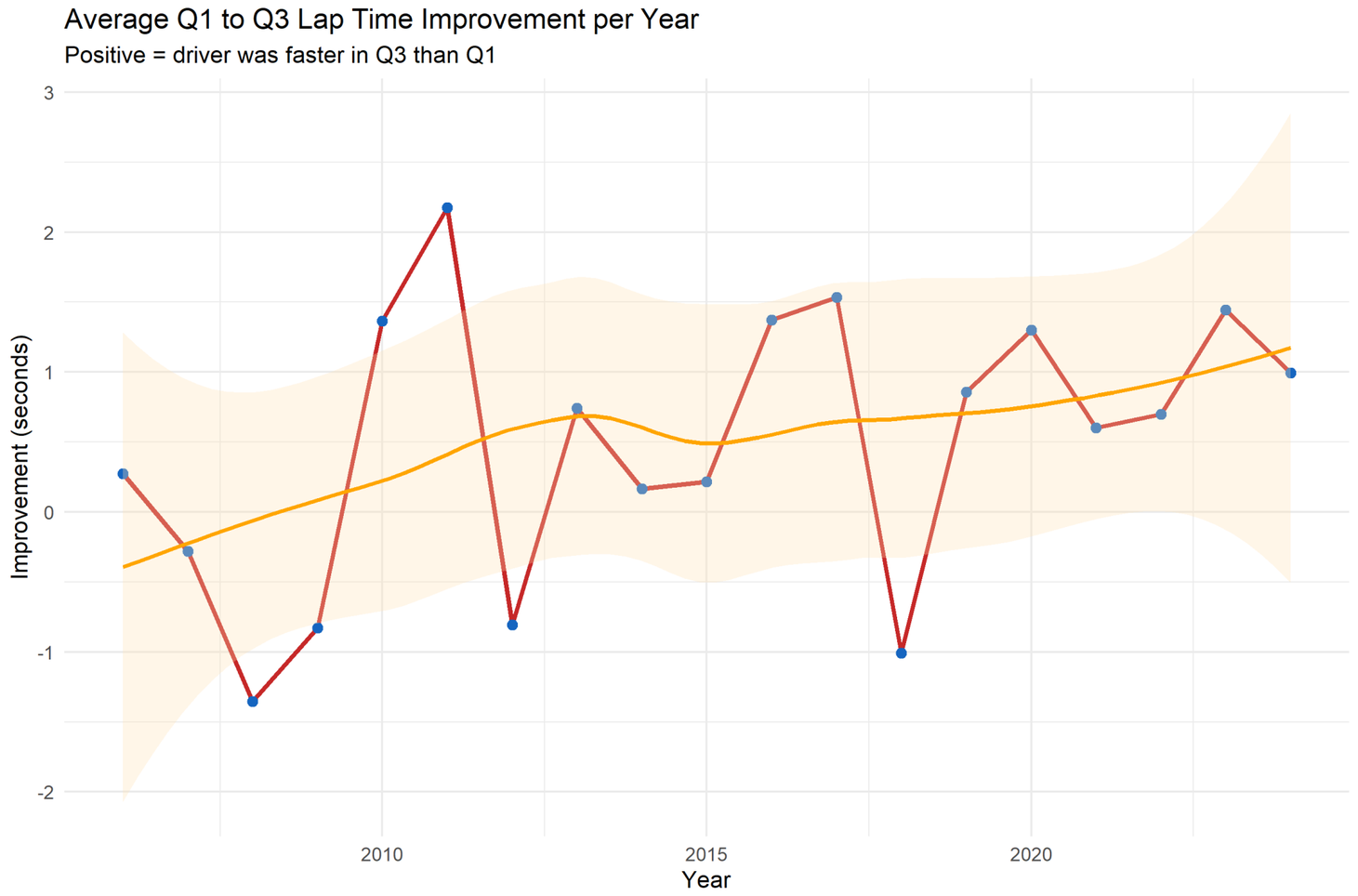
UNCERTAINTY

Bars show means; error bars show one standard deviation, not confidence intervals.

CAUTION

The measurement is the dataset's recorded duration. Do not equate it with stationary wheel-change time.

Qualifying: Q1 to Q3 time change



DEFINITION

Improvement = Q1 lap time minus Q3 lap time. Positive values mean a faster Q3 time.

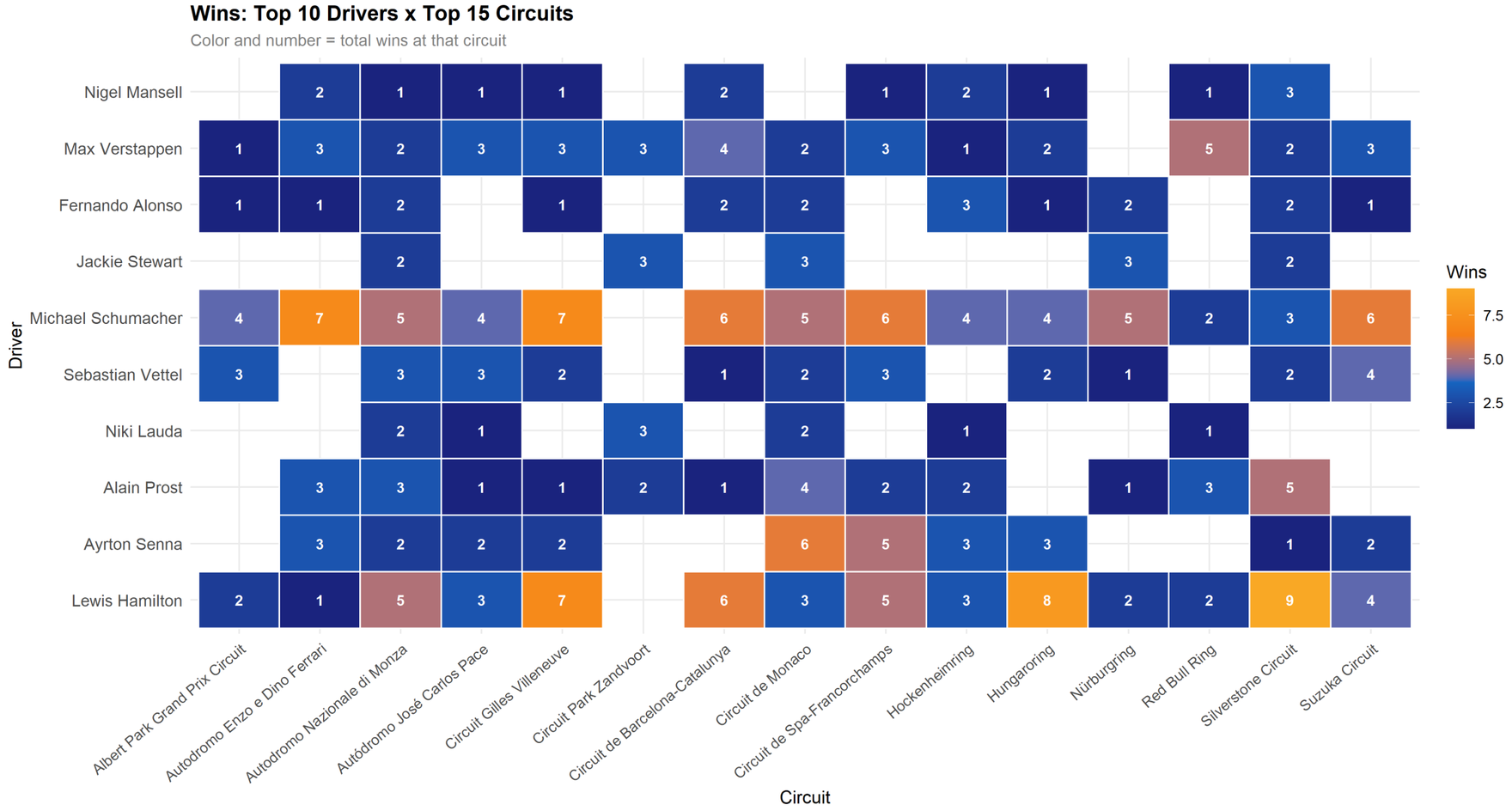
POPULATION

Only records with both Q1 and Q3 times are included. This is a selected group of qualifying participants.

INTERPRETATION

The smooth curve summarizes the observed sample. Format, track and weather changes limit historical comparisons.

Wins across drivers and circuits



SELECTION

The top 10 drivers by wins and the 15 most frequent winning circuits are selected from the input data.

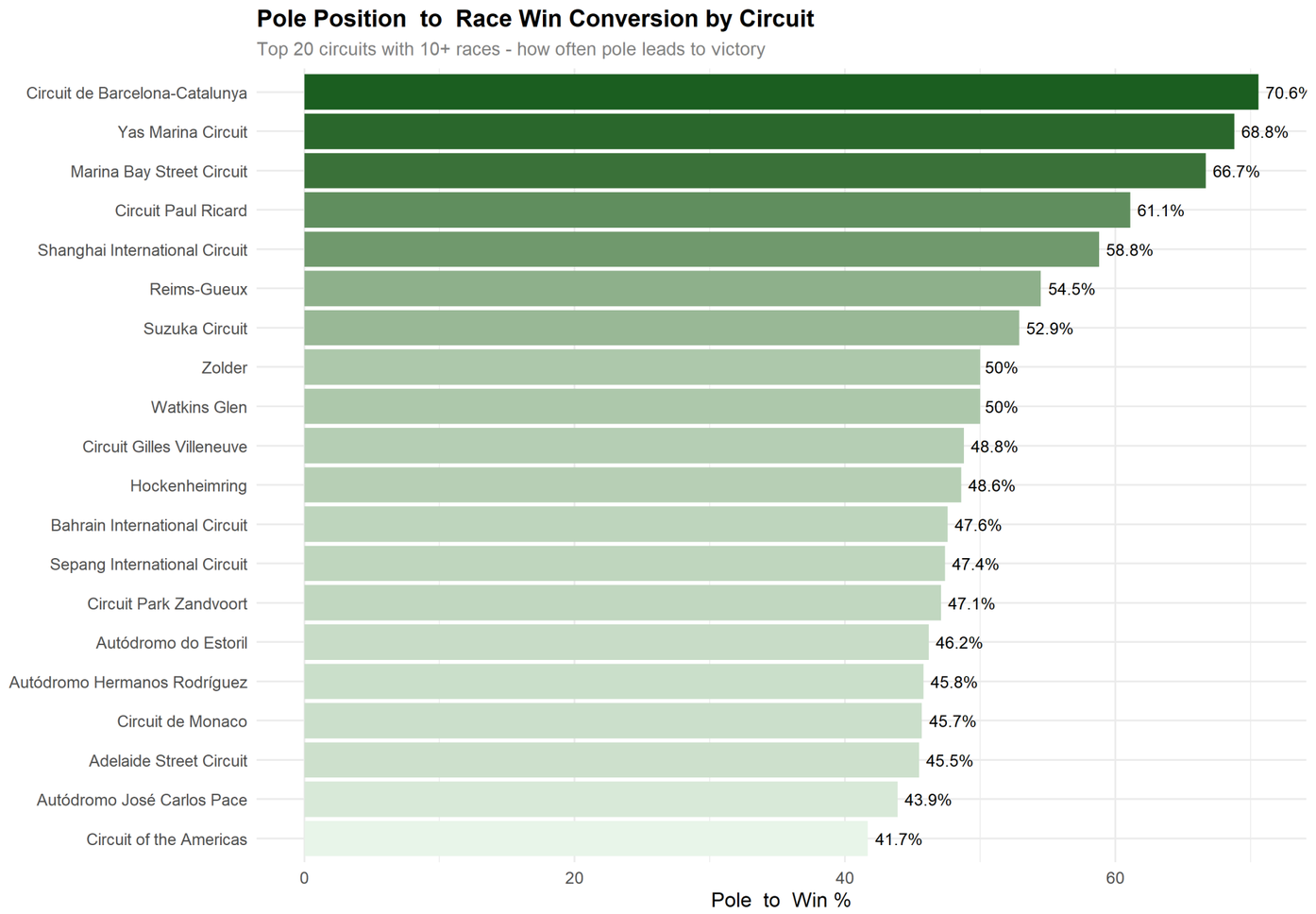
READING

Each filled cell counts wins. Blank cells indicate no winning observation for that selected pairing.

LIMIT

Totals reflect opportunity as well as performance; they are not normalized for starts or career length.

How often does pole become a win?



DEFINITION

Wins from grid position 1 divided by the number of observed races at each circuit.

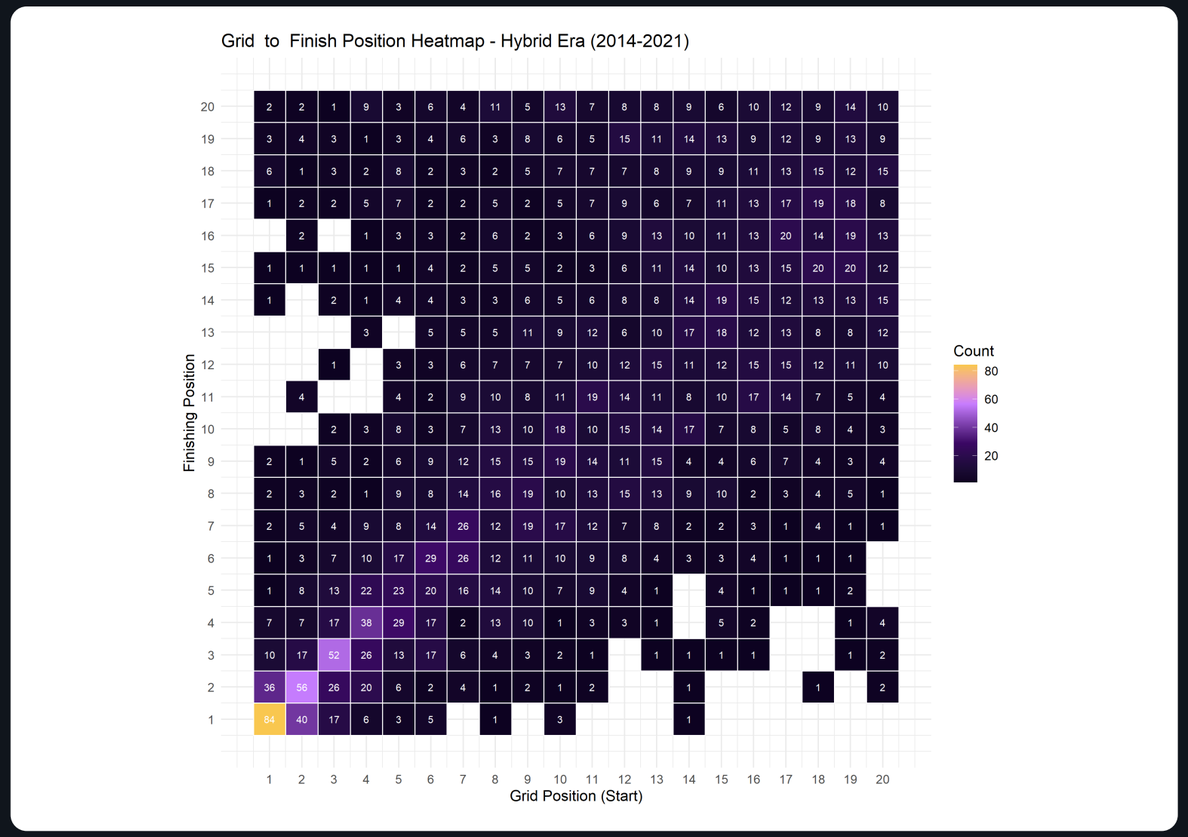
SELECTION

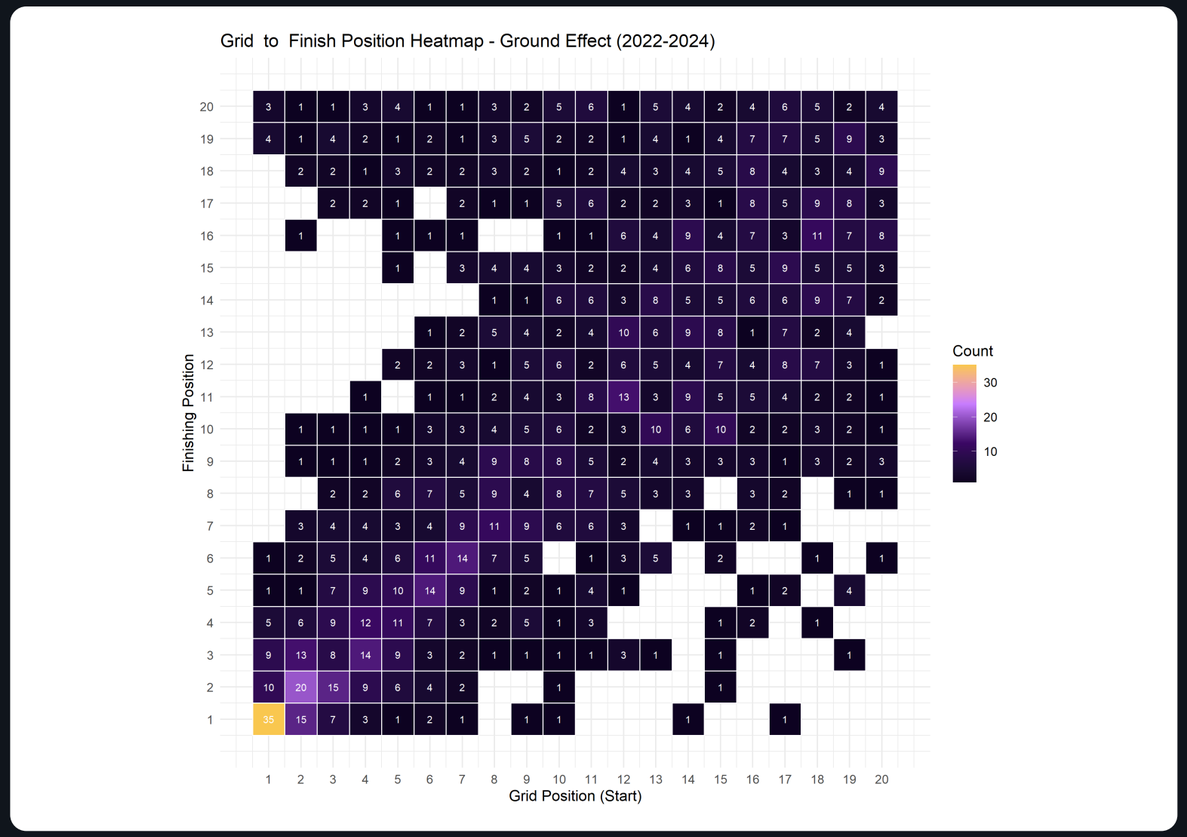
Circuits with at least 10 races; the 20 highest percentages are shown.

BOUNDARY

This is an observed historical proportion, not a causal effect or a forecast. Starting-grid exceptions affect interpretation.

Starting and finishing positions across eras





Cells count result rows by starting and finishing position (positions 1-20). These panels show two selected eras; the script exports all six. Raw counts depend on era length, calendar size and field size.

Findings, limitations and sources

Performance is multidimensional

The plots describe outcomes, geography, qualifying, pit stops and team connectivity. No single summary measures overall driver quality.

Validation checks consistency

The revised run preserves all 26,759 result rows and checks ID uniqueness, graph endpoints, degree totals, outcome totals and numeric bounds.

Historical comparisons need context

Rule changes, shared-car records, career length and measurement conventions remain relevant. Optional circuit-type charts require an explicit mapping.

Data: Rohan Rao, *Formula 1 World Championship (1950-2024)*, Kaggle version 24, updated 29 Jan 2025; retrieved 08 Nov 2026. Kaggle lists the data as CC0.

https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2020

Charts regenerated from the revised R script. Cover artwork is original vector geometry. Presenter: Irisbaev Nuriddin.